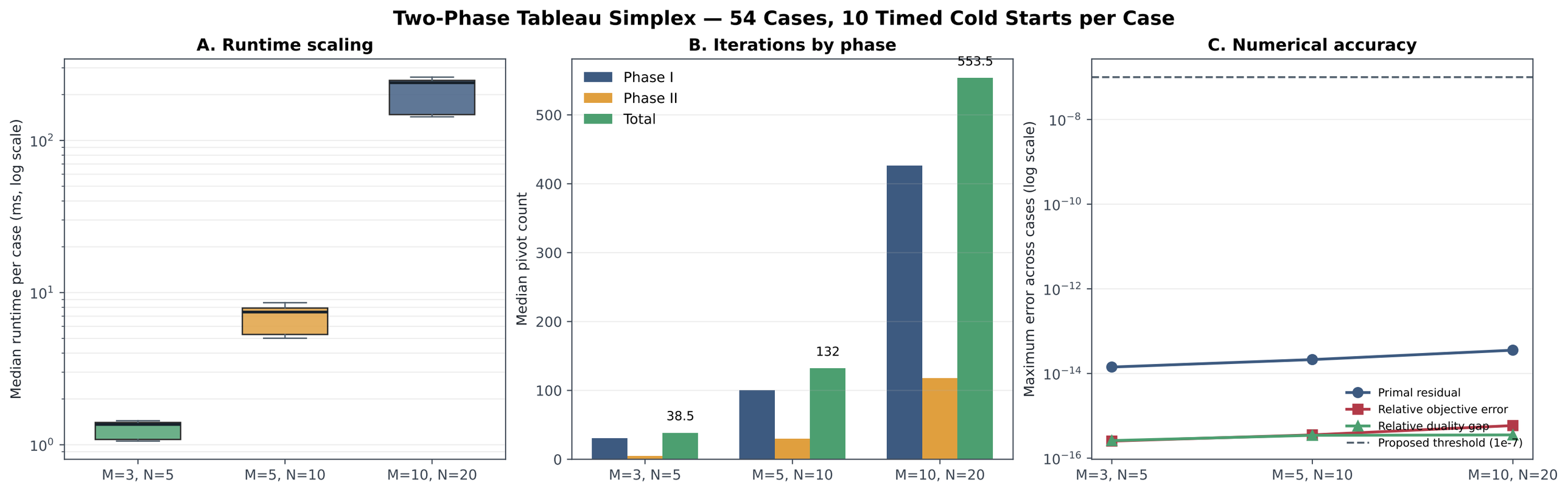
Two-Phase Tableau Simplex — 54 Cases, 10 Timed Cold Starts per Case
A. Runtime scaling
Median runtime per case (ms, log scale)
10^2
10^1
10^0
M=3, N=5
M=5, N=10
M=10, N=20
B. Iterations by phase
Median pivot count
Phase I
Phase II
Total
0
100
200
300
400
500
38.5
132
553.5
M=3, N=5
M=5, N=10
M=10, N=20
C. Numerical accuracy
Maximum error across cases (log scale)
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
Primal residual
Relative objective error
Relative duality gap
Proposed threshold (1e-7)
M=3, N=5
M=5, N=10
M=10, N=20